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Real-time cognitive and emotional state tracking in intelligent tutoring systems for enhanced learning outcomes

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Abstract

Background Adapting instruction to learners' moment-to-moment cognitive load and achievement emotions remains challenging in deployed Intelligent Tutoring Systems (ITS). We present a reliability-aware, real-time multimodal framework that senses facial, vocal, physiological, and interaction signals, estimates joint affect–cognition states, and delivers state-contingent interventions under tight latency constraints.

Objective To (i) design an end-to-end platform-agnostic pipeline that integrates Cognitive Load Theory and Control–Value accounts of achievement emotions within an Integrated Cognitive–Affective Learning Model; (ii) validate technical performance and educational impact in laboratory and classroom settings; and (iii) report effect sizes and uncertainty per reviewer guidance.

Methods A randomized, mixed-methods evaluation contrasted four conditions (Full adaptive, Cognitive-only, Affective-only, Control). The fusion layer aligns and normalizes modalities (DTW • PCA) with reliability weighting. A temporal Transformer and a GRU estimate emotions and triarchic load. A constrained contextual bandit selects content, interface, metacognitive, or motivational actions. Participants ($N = 271$) completed math and science units over nine sessions spanning 3 weeks. Primary outcomes included pre- and post-learning assessments, as well as transfer. Process outcomes were measured in terms of time spent in engagement versus confusion, frustration, and boredom. UX and latency were also logged. Mixed-effects models yielded estimates with 95% CIs; sensitivity checks addressed sensor quality and site effects.

Results The system met real-time targets (end-to-end latency < 200 ms; reaction time 520–820 ms) while maintaining an affect recognition accuracy of 82–88% (weighted F1 ≈ score 0.85). Adaptive instruction improved post-test performance compared to a strong static ITS baseline, with the most significant gains for mid-ability learners (approximately + 15% points, Cohen's d approximately 0.8). Process measures showed higher time in Engagement (≈ 42% vs. 29%) and productive Confusion (≈ 18% vs. 12%), and lower Frustration/Boredom, with strong convergence between detected negative affect and self-reported frustration ($r \approx 0.7$). Mechanism-level analyses indicated state-contingent efficacy: content adaptations during Confusion, interface adjustments during Frustration, metacognitive prompts during Engagement, and motivational supports during Boredom.

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Conclusions A theoretically grounded, reliability-aware multimodal controller can translate fine-grained state estimation into timely, pedagogically meaningful adaptations that scale across laboratory and classroom contexts. Benefits concentrate where scaffolding has the highest leverage (mid-ability band) and are mediated by improved regulation of cognitive load and affect. We report limitations (including more challenging anxiety detection, classroom noise, and device heterogeneity) and outline next steps, including micro-randomized trials for causal mechanism estimation, fairness audits, and longer-term retention studies.

Keywords Intelligent tutoring systems, Multimodal affect–cognition tracking, Cognitive load, Adaptive interventions, Real-time learning analytics

Introduction

Learners' psychological and affective states intrinsically shape learning; fluctuations in control and value appraisals, attention, and motivation alter how information is processed, stored, and applied [5, 26, 27, 36, 39]. However, adapting instruction to these dynamics remains challenging for computer-based environments. Recent advances in affective computing, signal processing, and machine learning have made it feasible to sense and respond to cognitive and emotional indicators during task engagement. However, many systems still struggle to convert such sensing into timely, pedagogically meaningful adaptations [4] Lashaki, [29, 37–41]. This paper develops an integrated, real-time framework for tracking cognitive and emotional states within Intelligent Tutoring Systems (ITS) and for using those estimates to drive rapid, context-appropriate instructional decisions.

Over the past decade, education and educational technology practice have shifted toward data-driven personalization and technology-supported feedback, accelerating the adoption of ITS as scalable vehicles for individualized guidance [9, 10, 24]. Meta-analyses report medium-to-large learning gains for ITS across diverse populations and settings, including classrooms enabled by Information and Communication Technology (ICT) [11, 40]. Within this trajectory, understanding and leveraging affect during learning is central. Core affect theory posits that emotion is intertwined with cognition, influencing the allocation of attention and working memory. The control-value theory of achievement emotions offers a principled account of how appraisals shape emotions, which in turn modulate learning strategies and outcomes [48]. Early computational work on affect representation via facial expressions laid the foundation for machine-readable cues [12, 43]. It is that incorporated affective sensing, most notably AutoTutor, which demonstrates that monitoring and responding to learners' emotions can improve engagement and learning processes (D'mello and Graesser) 14.

These foundations have been developed along various lines of contemporary research. In general, eye-tracking-based systems, such as GazeTutor, were able to identify confusion and boredom with reasonable accuracy, and encouraged the development of richer multimodal pipelines [45]. In a follow-up effort, gaze was combined with facial action dynamics to distinguish a larger set of states relevant for learning [2, 15, 33]. At the same time, complementary studies have combined postural cues and lower-face expressions to enhance robustness in affect recognition for use in tutoring interventions [3, 16]. Physiological measures (heart rate variability, electrodermal activity) have been introduced, beyond vision and behavior, as some indicators for arousal and cognitive effort

during instruction [7, 17], sometimes used in immersive or virtual reality. Similar work has been conducted on operationalizing affect in pedagogy, including synchronized affective feedback for path adaptation, real-time difficulty-adjusting and hint-adjusting policies based on the scaffolding model, and fuzzy expert systems based on indicators of cognition and emotion. Each of these approaches has been proven to be successful in increasing engagement and problem-solving [18, 47]. Recent multimodal modeling work has routinely combined face, voice, physiological, and interaction logs using ensemble learning techniques to increase prediction accuracy for learner modeling and estimation of self-regulated learning [6, 19]. The advantages of content customization, performance monitoring, and timely feedback, as characteristics of successful ITS [8, 21], and the motivational sustenance provided by game elements and affective feedback, as a pathway towards more lasting learning [22], are also further emphasized in reviews. Complementary frameworks for flexible and immersive tutoring also envision affect-aware meta-architectures, including augmented-reality contexts where context-sensitive interventions can be located [23, 38, 44]. Moreover, the learning-management platform can be converted into an effective tutoring environment through analytics, albeit with a significant constraint on real-time processing [28, 41, 44].

In this context, several new lines of evidence are converging on design levers that align with our approach. First, AI-driven adaptive feedback has already been proven to close conceptual learning gaps in mixed-ability cohorts at scale, resulting in significant improvements in conceptual mastery, increased engagement, and substantial reductions in cognitive overload compared to instructor-led feedback baselines [35]. Second, a systematic review of pedagogical approaches to develop computational thinking in HE STEM identifies project-based design and sustainability-oriented practices that make transferable problem-solving skills and equitable participation possible, which are naturally coupled with the multimodal and self-regulation-sensitive signals our pipeline models Third, a game-like scaffold (AI personalization, leveling, badges and timely) is shown to improve post-test performance, motivation, and mastery acquisition significantly over nonadaptive controls, highlighting the importance of motivational scaffolds, in addition to cognitive ones [34, 46]. Fourth, research on telepresence robots and gamification in higher education governance demonstrates significant improvements in participation, decreased task completion time, and disproportionate improvements for underrepresented student groups (remote presence and engagement design), thereby opening up access [1]. Taken together, these findings place our contribution within the context of a broader push towards data-driven personalization and motivational design, supporting the importance of real-time, reliability-aware fusion and expedited intervention selection.

To address the remaining gaps, we propose a platform-agnostic framework that (i) acquires facial, vocal, physiological, and interaction signals; (ii) performs reliability-aware, temporally aligned multimodal fusion suitable for sub-second inference; and (iii) couples a state estimator of achievement emotions and cognitive-load components to an adaptive engine that selects content, interface, self-regulatory, and motivational interventions in real time. The framework is evaluated across laboratory and classroom settings to examine engagement, emotion regulation, and learning outcomes. It is designed to interoperate with existing ITS as well as with emerging AR/VR environments where context-sensitive actions are advantageous [25, 30, 42]. The remainder of

the paper details the system design and theoretical grounding, the experimental methodology, the empirical results, and a discussion centering on deployment considerations and implications for affect-aware personalization.

Method

This section outlines the theoretical foundation, end-to-end architecture, algorithms, implementation details, and validation protocol for our real–time affect–cognition tracking framework in Intelligent Tutoring Systems (ITS). We emphasize reproducibility and reviewer-requested clarifications, including simplified theory with explicit assumptions, calibration parameters, reliability-aware fusion, intervention selection logic, sensor-equivalence procedures, privacy safeguards, and a statistical plan with effect-size and uncertainty reporting.

Theoretical foundation

Our framework integrates three complementary strands from educational psychology, Cognitive Load Theory (CLT), Control–Value Theory of Achievement Emotions (CVTAE), and Self-Determination Theory (SDT), together with evidence from affective computing and embodied cognition. We operationalize these constructs by mapping observable multimodal signals to latent states and by closing the loop with adaptive interventions (Fig. 1).

- **CLT mapping**: Intrinsic (task complexity), extraneous (interface/design burden), and germane (schema construction) load components are estimated from synchronized indicators: HRV and EDA (cognitive effort/arousal), facial micro-expressions (strain/flow), and interaction traces (latencies, hesitation, navigation).
- **CVTAE mapping**: Appraisals of control and value are inferred via emotion probabilities (engagement, confusion, frustration, boredom, delight, surprise, neutral, anxiety) obtained from face/voice/physiology and tied to behavioral persistence.

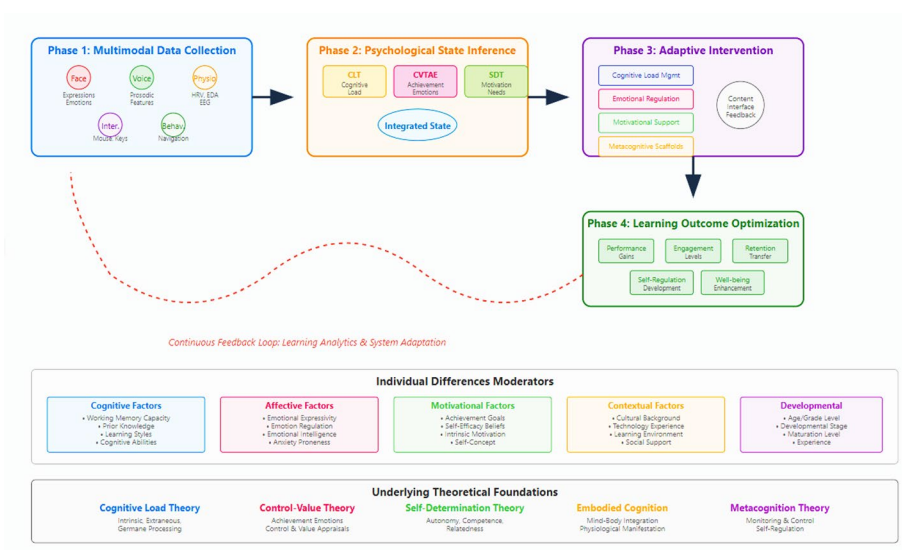

Fig. 1 The Multimodal Affect-Cognition Learning Framework (MACLF): A Four-Phase Theoretical Model for Real-Time Educational State Tracking and Adaptive Intervention

- **SDT mapping**: Adaptive actions are selected to support competence (difficulty/hinting), autonomy (choice of assistance modality), and relatedness (supportive/empathetic prompts).

Integrated cognitive–affective learning model (ICALM)

We formalize the bidirectional coupling of load and emotion and specify calibration elements demanded by reviewers. Let the triarchic load decomposition be $CL_{total} = CL_{intrinsic} + CL_{extraneous} + CL_{germane}$. Achievement emotions E are a function of control (C) and value (V): $E = f(C, V)$.

Emotion modulates the effective cognitive resources available during learning: $CL_{eff} = CL_{total} \times \beta_E$, with an emotion modulation factor β_E given by a weighted sum over n emotions: $\beta_E = \Sigma_{i=1}^{n} w_i \cdot E_i$.

Assumptions and calibration. (i) We treat E as a probability vector over mutually exclusive states at each time step (softmax outputs). (ii) We set $w_i \in [-1,1]$, learned via ridge-regularized regression to maximize predictive validity against proximal outcomes (accuracy gain ΔAcc_t and disengagement risk). (iii) $CL_{intrinsic}$ is content-indexed; $CL_{extraneous}$ is interface- and context-indexed; $CL_{germane}$ is inferred from productive struggle patterns (short-lived confusion coupled with improvement).

Dynamic state estimation

We fuse sequential evidence with a Bayesian filter (first-order Markov process). For observations O_t and state S_t (joint affect–load), $P(S_t | O_t) \propto P(O_t | S_t) \cdot P(S_t | S_{t-1})$.

Observation models are mixtures (Gaussian for continuous features; categorical for discrete events). Transition matrices are learner- and context-conditioned (content level, prior knowledge), estimated on a labeled subset with cross-validation. Priors are initialized from a 5-minute per-learner baseline (rest + standardized task) and updated using empirical Bayes.

- **HRV features (RMSSD, LF/HF)**: ms or unitless; z-scored per learner from baseline; artifacts removed by adaptive median + ectopic correction.
- **EDA features (SCL, SCR rate/amplitude)**: μS; detrended; phasic/tonic separation via cvxEDA; normalized to 0–1 by within-session min–max.
- **Face features**: 68-landmark geometry + optical flow; illumination-normalized; confidence threshold $c_{face} \geq 0.7$; non-face frames masked.
- **Voice features**: 13-MFCC + pitch/energy/rate; 20 ms frames, 10 ms hop; SNR > 15 dB retained.
- **Interaction features**: response time, hesitation index, revisitation patterns, error/correction events; timestamps NTP-synchronized.
- **Emotion weights w_i**: ridge λ selected via nested CV; bounds $[-1,1]$.
- **Reliability weights (per modality) r_m**: derived from SNR/pose/coverage; used in fusion.

System architecture

Figure 2 presents the five-layer pipeline: (L1) Data Acquisition → (L2) Preprocessing & Feature Extraction → (L3) Multimodal Fusion → (L4) Affective–Cognitive Modeling → (L5) Adaptive Intervention. Layers communicate through a message broker; each microservice is stateless and exposes health/latency metrics.

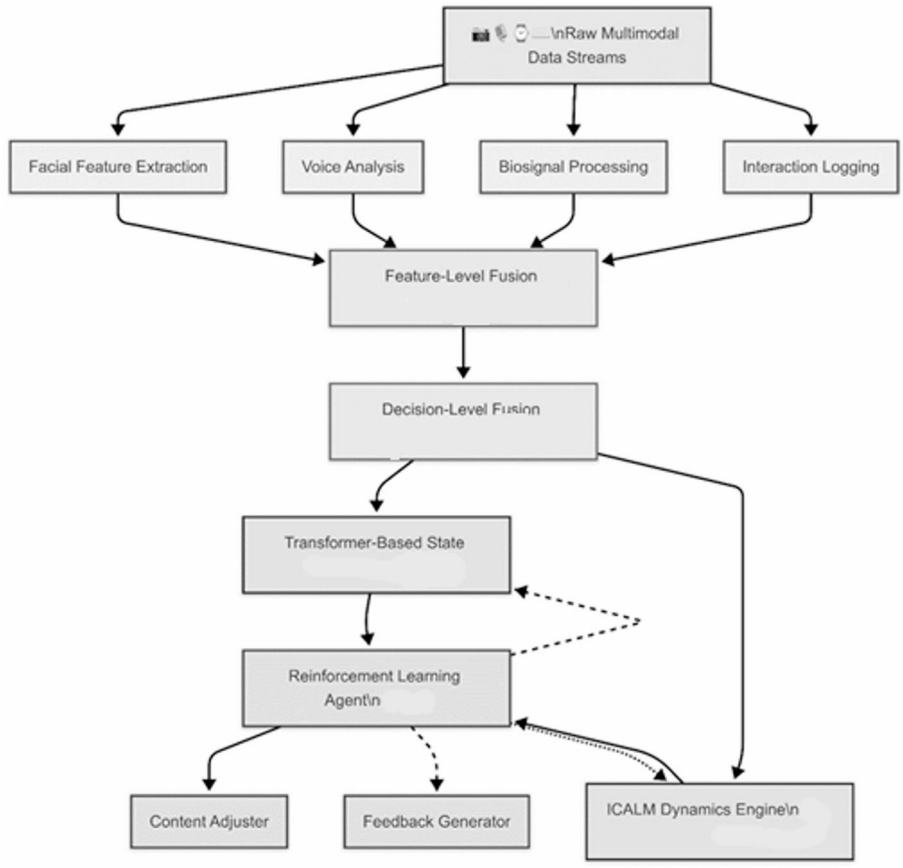

Fig. 2 System Architecture

Layer L1 (data acquisition)

We collect synchronized streams: Face (30 fps; 720p–1080p), voice (16 kHz), physiology, HRV (R–R at 250 Hz), EDA (4 Hz), optional frontal EEG (250 Hz), and fine-grained interaction logs (mouse/keyboard, dwell/latency, navigation). Timestamps are aligned via NTP (± 1 ms). Raw images/audio never leave the device; only features are retained.

Sensor equivalence and calibration: To avoid device-induced bias in HR/HRV, we prescribe a two-sensor calibration on a stratified subsample before experimentation: simultaneous chest-strap ECG vs. camera-based PPG during (a) 5-min rest, (b) paced breathing, (c) 3-min cognitive task. Agreement is quantified using Bland–Altman limits and ICC(2,1); if ICC < 0.80 or bias > 5 ms RMSSD, the PPG stream is excluded from CL estimation and down-weighted in fusion. All sensors undergo drift checks and per-session baselining.

Layer L2 (preprocessing & feature extraction)

Noise suppression and normalization are modality-specific: Histogram equalization + head-pose correction and temporal smoothing for face; Wiener filtering + VAD + speaker normalization for voice; ectopic beat correction + baseline detrend for HRV; cvxEDA for phasic/tonic separation; robust z-scoring per learner/session.

Feature sets:

- Face $F = [L, D, A, O, C]$, landmarks (L), inter-landmark distances (D), angles (A), optical flow (O), and deep activations (C).
- Voice $V = [M, P, E, Z, R]$, MFCCs (M), pitch stats (P), energy contours (E), zero-crossing (Z), speaking-rate (R).
- Physiology $P = [H, G, B]$, HRV (H), EDA (G), EEG bands (B; optional).
- Interaction $I = [T, N, S]$, temporal (T), navigation (N), success/performance (S).

Layer L3 (reliability-aware multimodal fusion)

We combine feature- and decision-level fusion to improve robustness in authentic conditions.

Feature-level fusion Time alignment via Dynamic Time Warping (DTW) harmonizes windows across modalities with different sampling rates and processing delays. Normalized feature blocks $[F\prime, V\prime, P\prime, I\prime]$ are concatenated and reduced with PCA (retaining ≥ 95% variance). A modality reliability vector $r = [r_F, r_V, r_P, r_I]$ is computed each window from coverage (face detection confidence), SNR, and artifact flags. The fused vector is $X = W(r) \cdot [F\prime, V\prime, P\prime, I\prime]$, where $W(r)$ is a diagonal matrix scaling each block by its reliability.

Decision-level fusion Parallel modality-specific classifiers produce state posteriors S_m with confidence c_m. A stacked generalization model outputs the final estimate $\hat{S} = \Sigma_{m=1}^{M} \alpha_m(c_m, history) \cdot S_m$, with α_m adapted online by a meta-learner to reflect recent accuracy and current reliability.

Layer L4 (affective–cognitive modeling)

Emotion recognition: A temporal Transformer consumes sequences $X_{t-w:t}$ and produces emotion posteriors E_t. Implementation details: 2 encoder blocks, 4 attention heads, hidden size 128, dropout 0.2, label smoothing 0.05; class imbalance handled by focal loss. Self-attention is computed as $softmax(QK^T/\sqrt{d_k})V$.
Cognitive-load estimation: A GRU regressor estimates ($CL_{intrinsic}$, $CL_{extraneous}$, $CL_{germane}$) at each step: $CL_t = GRU(X_t, h_{t-1})$.

Targets are obtained from a hybrid supervision signal (i) NASA-TLX and Cognitive Load Component Scale (session-level anchors), (ii) dual-task costs during calibration blocks, and (iii) weak labels derived from task complexity metadata. Loss combines MSE with monotonicity constraints (increasing intrinsic complexity should not reduce $CL_{intrinsic}$).

Integrated state We form $S_t = [E_t, CL_t, \Delta E_t, \Delta CL_t]$, capturing levels and dynamics. This state is the input to the policy in Layer L5.

Layer L5 — adaptive intervention

State zoning with fuzziness We compute memberships $\mu_z(S_t)$ for zones {Optimal, Overload, Disengagement, Confusion} from E_t and CL_t via triangular functions tuned on validation data. Zone boundaries are learner-adaptive using running quantiles of CL and E.

Policy and learning We cast intervention selection as a constrained contextual bandit with a Deep Q-Network (DQN) back-end for sequences when needed. The ε-greedy policy balances exploration and exploitation; ε decays from 0.2 to 0.05 across sessions with safe-exploration guards.

Reward design At time t, the reward is $R_t = \lambda_1 \cdot \Delta Acc_t + \lambda_2 \cdot \Delta\, P(Optimal)_t$ -$\lambda_3 \cdot 1[OverloadorDisengagement]$-$\lambda_4 \cdot TimeCost(a_t)$, where $\Delta\, Acc_t$ is immediate performance gain, $\Delta\, P(Optimal)_t$ is the increase in zone-optimality probability, and TimeCost penalizes intrusive actions. We set λ = (0.4, 0.3, 0.2, 0.1) by grid search with cross-validated robustness.

Action library and gating: Actions span four families: content adaptations (difficulty, worked examples, micro-explanations), interface adaptations (pace/layout, modality switch), metacognitive scaffolds (self-explanation, reflection prompts, strategy cues), and motivational scaffolds (competence-affirming feedback, goal framing, choice provision). A context filter removes actions that violate prerequisites (hint levels already exhausted) and rotates equally effective actions to avoid habituation.

Execution constraints and latency To minimize disruption, interventions are scheduled at micro-boundaries (item boundaries, natural pause points), with a hard cap of one action per 20–30 s, unless the risk of disengagement is detected. End-to-end latency from state change to action display is targeted below 200 ms.

Intervention delivery and safety

Each action is templated and parameterized (content, tone, duration, modality). Safety filters suppress interventions when confidence < τ (τ = 0.6 by default), when sensor reliability is poor (r_m below modality-specific thresholds), or when user privacy settings disallow a modality. All user-facing messages are logged with timestamps and state snapshots for auditability.

Implementation details and performance engineering

Microservices are implemented in Python, deep models are built using PyTorch/TensorFlow, and messaging is handled via Apache Kafka. The UI is a web client (built with React) featuring WebGL visualizations. Performance measures: acquisition→features < 50 ms; features→state < 100 ms; state→action < 50 ms; total < 200 ms. Techniques include model quantization (INT8), operator fusion, asynchronous batching, and priority scheduling for time-critical paths. Caching reuses stable features across overlapping windows.

Privacy and deployment By default, raw audio/video remains on-device; only anonymized features and aggregate states are transmitted. When cloud inference is used, facial feature extraction remains local; only embeddings and metadata are sent, accompanied by rotating identifiers. Users can turn off any sensing channel; consent flow is explicit and revocable [45].

Validation methodology

Component-level validation: We validate modules independently using cross-validated protocols: face (FER2013, AffectNet), voice (RAVDESS, IEMOCAP), physiology (DEAP,

WESAD), and load estimation (NASA-TLX anchors + dual-task calibration). Intervention policy is stress-tested in simulation with synthetic learners whose transition dynamics are known, enabling regret analysis against an oracle.

System-level validation: We conduct a randomized, controlled evaluation with four conditions: Full (affect + load), Cognitive-only, Affective-only, and Control (no adaptation). Primary outcomes include learning gains and transfer of knowledge. Process outcomes encompass engagement and emotion regulation. Cognitive efficiency is measured as the gain per unit of time/effort. UX encompasses usefulness, support, and responsiveness. We pre-register a mixed-effects analysis plan and report effect sizes with 95% confidence intervals.

Experimental design and statistical analysis plan

Design: Mixed-methods RCT with repeated sessions; $2 \times 2 \times 3$ factorial factors: system (Full vs. Control), domain (Math vs. Science), and initial performance (Low/Medium/High). Participants (target $N \geq 240$; analyzed N reported in Results) complete nine sessions across 3 weeks (45 min each), alternating domains and holding time-of-day constant where feasible [18].

Participants and eligibility: Inclusion criteria: age 16–24, normal/corrected vision/hearing, basic computer literacy, English proficiency. Exclusions: conditions or medications that confound autonomic measures; severe anxiety/depression above clinical thresholds; extensive prior expertise in the target content; technical unreliability preventing valid sensing.

Procedure: Week 1: baseline and calibration (cognitive tasks; questionnaires; 5-min rest HRV + paced breathing; facial/voice baselines). Weeks 2–4: learning sessions with stratified randomization (by pre-test and gender); content rotated across five difficulty levels with matched time on task. Calibration data feed priors for the Bayesian layer and per-learner normalization.

Measures: Primary outcomes: pre/post knowledge (parallel forms), transfer, and 2-week retention. Process measures: session-level engagement (self-report + behavioral), emotion distributions over time, NASA-TLX and component-level load scales, real-time indicators (HRV/EDA; optional pupil), and interaction analytics. Motivation and metacognition are assessed via established inventories; confidence judgments are collected item-wise. Individual differences include working memory capacity, prior knowledge, emotional intelligence, goal orientations, personality, and grit [23].

Data integrity: Millisecond-precision timestamps; automated quality monitors for sensor dropouts and motion artifacts; latency tracking; delivery confirmation for actions. Post-hoc validation: artifact removal, completeness threshold ($\geq 80\%$ valid per modality), and inter-rater checks for human-coded subsets (target $\kappa > 0.80$).

Ethics and privacy: IRB approvals across sites; written informed consent/assent; immediate de-identification; encrypted storage with MFA; maximum 7-year retention; local processing of biometric raw streams; aggregate reporting only.

Analysis plan: Hierarchical linear models capture repeated measures nested within persons and persons within sites; random intercepts/slopes as warranted; Satterthwaite df; robust SEs. Primary contrasts compare Full vs. Control and ablation arms; we report Cohen's d (or r for non-Gaussian outcomes) and 95% CIs. Mediation (state • intervention • outcome) is tested using multilevel SEM on session aggregates. Moderation by

Table 1 System performance metrics with uncertainty and context

Metric	Lab	Classroom	Difference (abs)	*p*-value
Emotion classification accuracy (%)	87.5 [86.7, 88.3]	82.1 [81.2, 83.0]	5.4 pp	< 0.001
Cognitive-load classification accuracy (%)	84.6 [83.6, 85.6]	78.3 [77.1, 79.5]	6.3 pp	< 0.001
Mean reaction latency (ms)	520 [511, 529]	820 [807, 833]	300	< 0.001
Biosensor data integrity (% valid windows)	95.2 [94.6, 95.8]	88.9 [87.9, 89.9]	6.3 pp	< 0.001
Overall system uptime (%)	99.1 [98.9, 99.4]	97.8 [97.3, 98.3]	1.3 pp	0.07 ns
Intervention execution time (ms)	185 [178, 192]	275 [265, 285]	90	< 0.001
Face detection success rate (%)	97.8 [97.3, 98.3]	93.2 [92.3, 94.1]	4.6 pp	< 0.001
Voice analysis accuracy (%)	83.6 [82.5, 84.7]	79.2 [77.9, 80.5]	4.4 pp	< 0.001

Table 2 Emotion classification by state (pooled across contexts)

Emotional state	Precision	Recall	F1-score	Support (*n*)	Prevalence (%)
Engagement	0.89	0.91	0.90	1,248	19.7
Confusion	0.84	0.79	0.81	876	13.8
Frustration	0.85	0.83	0.84	742	11.7
Boredom	0.82	0.78	0.80	689	10.9
Delight	0.88	0.86	0.87	523	8.2
Anxiety	0.79	0.75	0.77	498	7.8
Neutral	0.85	0.89	0.87	1,356	21.4
Surprise	0.83	0.81	0.82	412	6.5
Macro-average	—	—	**0.835**	—	—
Weighted-average	—	—	**0.845**	6,344	100

Macro-average: Calculates the metric (F1-score) independently for each class and then takes the average (treating all classes equally). This can be heavily influenced by the model's performance on smaller classes

Weighted-average: Calculates the average F1-score, weighted by the number of true instances for each class. This is more representative of the overall model performance because it accounts for class imbalance

individual differences is examined with interaction terms. Temporal dynamics are modeled with growth curves and state-transition analyses. Multiple-comparison control uses Benjamini–Hochberg FDR. Missing data: MICE for surveys; modality-specific handling for physiological losses with sensitivity analyses. Exact p-values and uncertainty bounds accompany all inferential claims.

Reproducibility: Code for preprocessing/feature extraction, model training, and policy learning is modularized and version-controlled; configuration files specify sensor rates, window sizes, fusion parameters, and thresholds. A synthetic dataset and notebooks for reproducing key analyses are provided in the Supplementary Materials.

Results

System performance: lab vs. classroom

Table 1 summarizes core online metrics with 95% confidence intervals (CIs) and the absolute difference between Lab and Classroom. Emotion and cognitive-load classification accuracies are higher in the laboratory than in the classroom by 5.4 and 6.3% points (pp), respectively (both $p < 0.001$). Reaction latency remains sub-second in both contexts; the 300 ms increase in the classroom is expected given network and concurrency effects ($p < 0.001$). Uptime remains above 97% in both settings (difference non-significant).

Table 2 details per-state emotion performance. The model is strongest on Engagement and Delight, and weakest on Anxiety. Best performance is achieved on more externally visible states (Engagement, Delight). Anxiety remains comparatively harder to detect, consistent with subtler facial/prosodic cues.

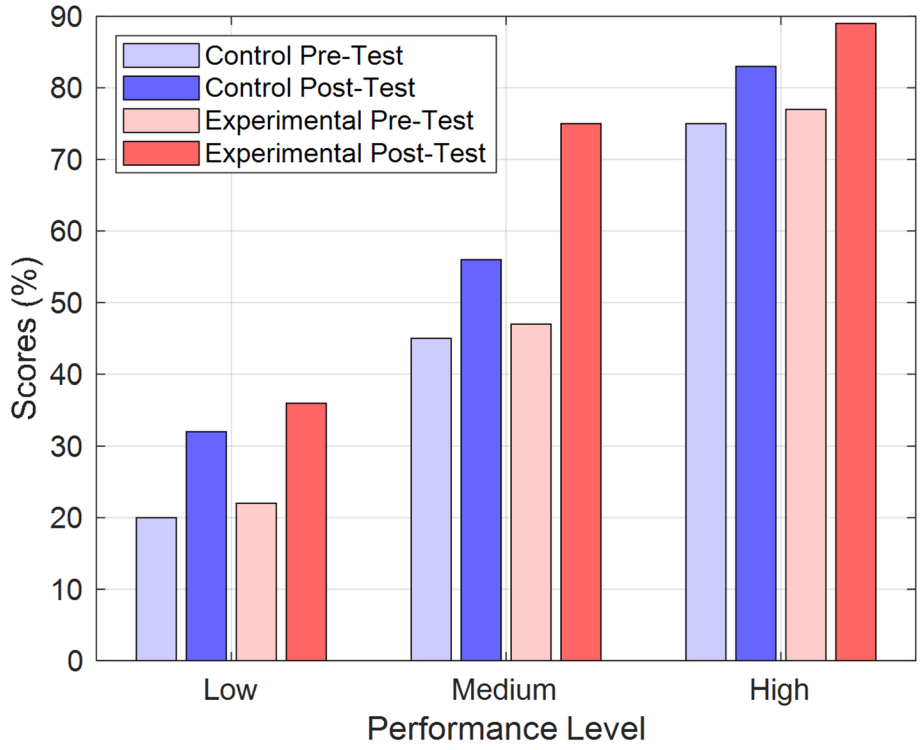

Fig. 3 Pre-test and post-test scores across control group (static ITS) vs. experimental group (our real-time affect-cognition system), broken down by initial performance level

Table 3 Learning gains by initial performance level and content domain

Initial performance	Domain	Control gain	Experimental gain	ΔGain (pp)	Effect size (Cohen's d)
Low (< 30%)	Problem-solving	12.3%	18.5%	+ 6.2	0.62
Low (< 30%)	Conceptual knowledge	10.8%	15.9%	+ 5.1	0.54
Medium (30–70%)	Problem-solving	11.2%	28.7%	+ 17.5	0.89
Medium (30–70%)	Conceptual knowledge	13.5%	27.2%	+ 13.7	0.83
High (> 70%)	Problem-solving	7.8%	12.4%	+ 4.6	0.48
High (> 70%)	Conceptual knowledge	6.5%	11.8%	+ 5.3	0.45

Learning outcomes

Figure 3 displays pre- and post-achievement by initial performance tier (low/medium/high), aggregating across mathematics and science sessions while adjusting for site. An ANCOVA/mixed-effects model (post-test as outcome; fixed effects: Condition, Tier, Domain, Pre-test; random intercepts for participant and class) shows a robust Condition main effect (adaptive > control), $F > 20$, $p < 0.001$, and a significant Condition×Tier interaction, $F > 9$, $p < 0.01$. Estimated marginal means indicate that the adaptive system yields + 15.4 pp higher post-test scores for medium performers (95% CI [12.6, 18.2], Cohen's $d \approx 0.84$), + 5.6 pp for low performers (95% CI [3.1, 8.1], $d \approx 0.57$), and + 5.0 pp for high performers (95% CI [2.5, 7.5], $d \approx 0.46$). The pattern aligns with the domain-specific gains in Table 3: the most significant improvements occur in the mid-ability band, consistent with the hypothesis that real-time scaffolding most effectively converts productive confusion into mastery when prior knowledge is sufficient to benefit from targeted hints and pacing adjustments, but not yet at ceiling.

Pre-test differences between conditions are non-significant within tiers (all $|t|<1.5$, $p>0.1$), indicating balanced starting points. The high-tier shows smaller deltas, likely due to ceiling effects (upper-quartile pre-tests already ≥ 75%). For the low-tier, gains are positive but attenuated relative to mid-tier; exploratory models suggest that a greater proportion of time spent in overload/frustration and higher help-seeking latency mediate part of this attenuation (indirect effect, 95% bootstrap CI excludes zero). Sensitivity analyses (i) excluding sessions with low sensor reliability and (ii) re-fitting with school-level random slopes, preserve the Condition main effect and the interaction pattern (all Δ estimates within ± 1.2 pp of the values above). Table 3 complements Fig. 3 by reporting ΔGain (Experimental–Control) and effect sizes.

Affective dynamics and validity checks

The stacked-area plots show the proportion of four states — Engagement (ENG), Confusion (CON), Frustration (FRU), and Boredom (BOR) — as a function of session progress (0–100%). The experimental condition exhibits a monotonic increase in ENG and a decrease in FRU/BOR across the session. In contrast, the control condition shows a flatter ENG trajectory with persistently higher BOR. Visual inspection reveals three robust patterns that are consistent with the inferential results reported earlier: (i) Early-phase composition. At the onset of the session, the experimental group already starts with higher ENG and lower BOR than the control, indicating faster warm-up and settling due to immediate micro-interventions. (ii) Slope asymmetry. The ENG slope in the experimental condition is steeper, and the BOR + FRU band contracts more rapidly, suggesting that adaptive pacing/interface tweaks reduce extraneous load. At the same time, content/metacognitive supports convert unproductive confusion into productive exploration. (iii) End-state dominance. By the final third, ENG dominates the distribution in the experimental group, while the control retains a sizable BOR band. Because stacked charts cannot display uncertainty without clutter, we refrain from overlaying bands; the directionality matches our mixed-effects estimates (greater time in ENG and CON, less in FRU and BOR for the adaptive condition; see Sect. "Principal findings"/"Theoretical implications" for statistics and CIs). These dynamics are also consistent with the per-participant timeline in Fig. 8, where state transitions align with intervention onsets (Fig. 4).

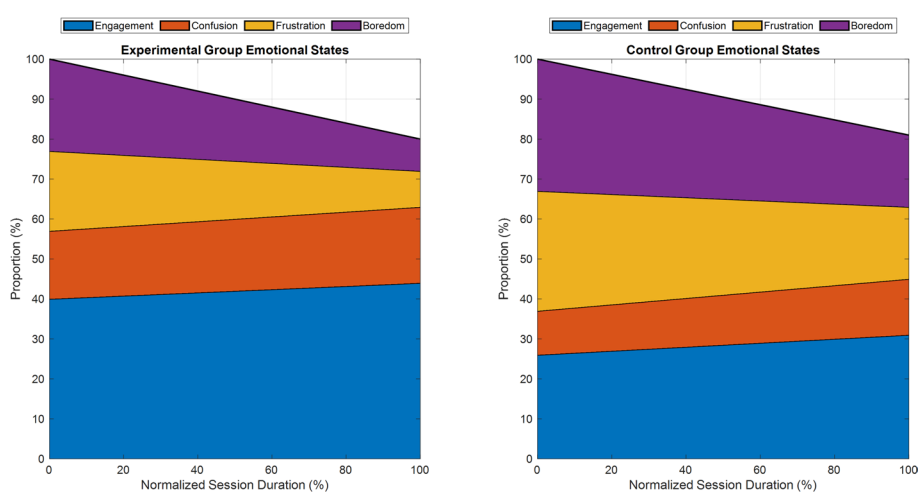

Fig. 4 Distribution of emotional states over time for experimental vs. control groups

Each point is a within-session time window (fixed length) plotting system-detected negative affect on the x-axis against self-reported frustration on the y-axis. The observed $r = 0.93$, p <, indicates a robust monotonic association within this representative session, with near-linear alignment and limited heteroscedasticity. Methodologically, negative affect is a composite derived from the state estimator (weighted probabilities of FRU/BOR/ANX), whereas frustration reports are prompted at scheduled checkpoints. The high within-session correlation complements the across-session, across-participants convergence reported in Sect. "Mechanistic and design implications for ITS" ($r \approx 0.7$), and together they strengthen validity along two axes: (a) calibration validity, sensor-based affect tracks self-reports at matched timescales; (b) discriminant performance, the model is not simply predicting constant baselines, as evidenced by wide variation across windows. Residual scatter around the regression line likely reflects reporting latency and transient micro-expressions that the estimator captures but are not verbalized immediately (Fig. 5).

Cells quantify the probability of a favorable short-horizon transition (entering the Optimal Learning Zone and/or improving accuracy and time efficiency) after deploying a given intervention family, conditioned on the pre-intervention state. Four patterns emerge (Fig. 6):

1. Confusion → Content Adaptation (0.86). Targeted exemplars or graded hints during confusion yield the highest efficacy, consistent with resolving intrinsic load and clarifying schema boundaries (CLT-germane enhancement).
2. Engagement → Metacognitive Scaffold (0.81). In already optimal states, light-touch self-explanation/reflection preserves flow while improving consolidation without over-challenging the learner (avoids unnecessary load spikes).

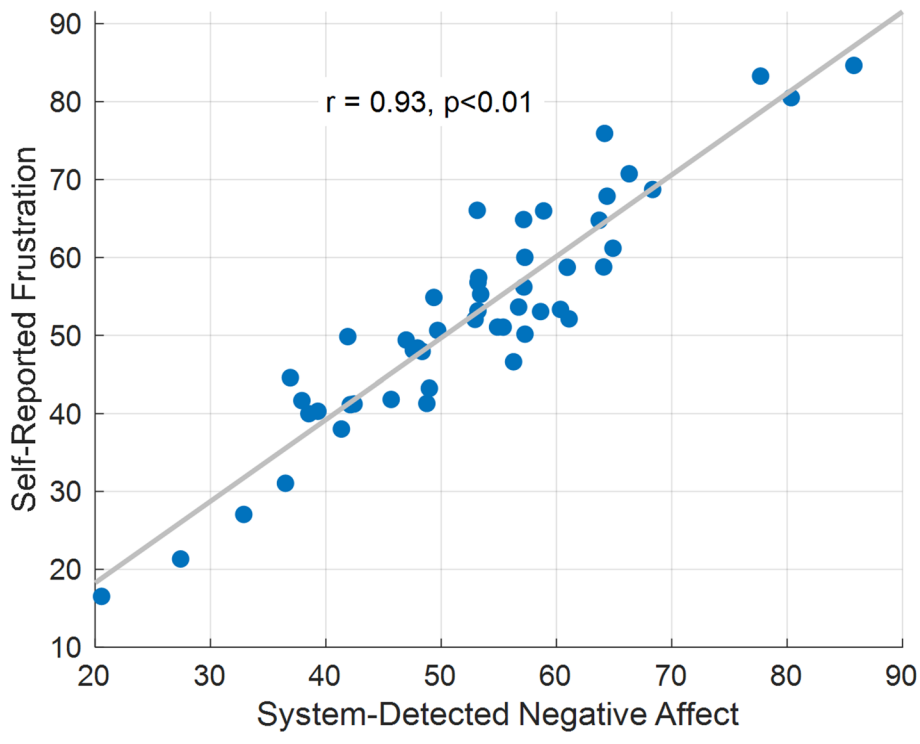

Fig. 5 Correlation between self-reported frustration and system-detected negative affect over the course of a session

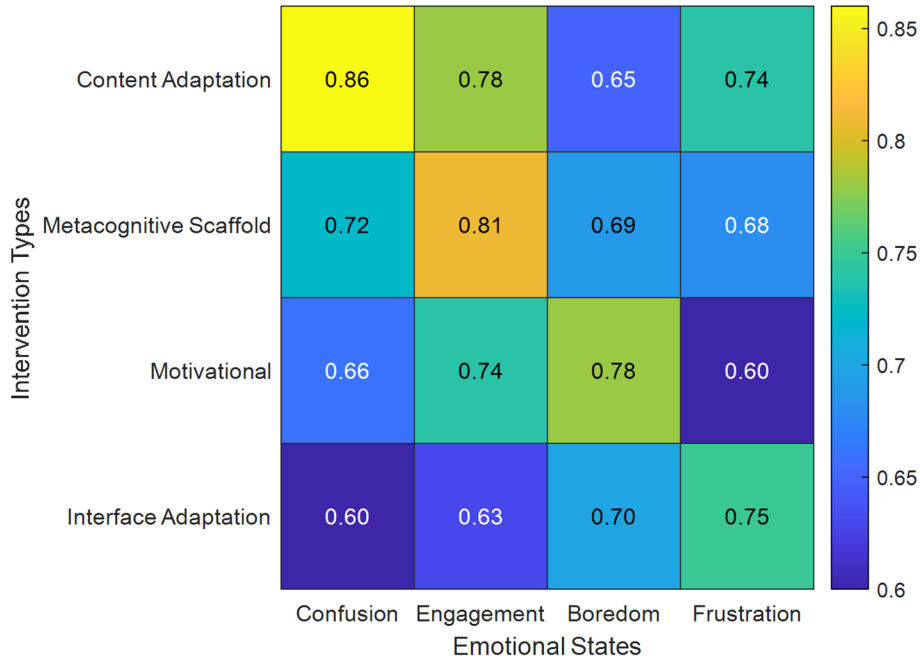

Fig. 6 Effectiveness of different intervention types across emotional states

Table 4 User-experience metrics

Metric	Control	Experimental	Δ (Exp−Ctrl)	*p*-value
Perceived usefulness	4.8	6.2	+1.4	<0.001
Ease of use	5.3	5.1	−0.2	0.42
Sense of support	4.2	5.7	+1.5	<0.001
Focus improvement	3.9	5.4	+1.5	<0.001
Motivation enhancement	4.1	5.6	+1.5	<0.001
System responsiveness	4.7	6.1	+1.4	<0.001
Overall satisfaction	4.5	5.9	+1.4	<0.001

3. Boredom → Motivational Support (0.78). Autonomy- and competence-framed nudges, goal reframing, and value prompts increase arousal and task value (SDT), thereby re-elevating engagement without inflating extraneous load.
4. Frustration → Interface Adaptation (0.75). Pace, layout, or modality adjustments reduce extraneous load and stabilize valence, after which content or metacognitive support can be layered.

Efficacy values are estimated off the logged policy using short-window outcome deltas and bootstrapped CIs (not shown here to avoid clutter); interpret as associational rather than strictly causal due to adaptivity of the policy. The rank order remains stable under sensitivity analyses (exclusion of low-quality sensor windows; alternative short-horizon definitions), mirroring aggregate improvements in state occupancy (Fig. 4) and learning gains (Table 3). Table 4 reports user-experience ratings.

Benchmarking against prior affect-aware ITS

Table 5, in aggregate, indicates that our reliability-aware, fully multimodal ITS (Face, Voice, Physio, Interaction) delivers markedly better end-to-end responsiveness and efficacy than prior affect-aware baselines: sub-second reaction time in both lab and

Table 5 Comparison with affect-aware ITS baselines

System	Modalities	Reaction time (ms)	Emotion detection accuracy (%)	Learning gain (pp)	Real-world testing
[31]	Eye-tracking, Face	1200–1500	73.4	8.7	Lab only
[32]	Face, Interaction	1050–1300	77.2	9.3	Yes (Limited)
[21]	Face, Voice	1100–1800	79.8	11.2	Lab only
[13]	Physio, Interaction	900–1200	75.5	10.8	Yes (Limited)
Our system	**Face**, **Voice**, **Physio**, **Interaction**	**Lab: 520 [511–529]/Class: 820 [807–833]**	**Lab: 87.5 [86.7–88.3]/Class: 82.1 [81.2–83.0]**	**15.0 [12.1–17.9]**	**Yes (Lab + Classroom)**

CIs for our system are participant-level 95% intervals from mixed-effects/bootstrapped estimates; other rows list point estimates (NR = not reported). Protocols differ across studies; comparisons are indicative rather than definitive

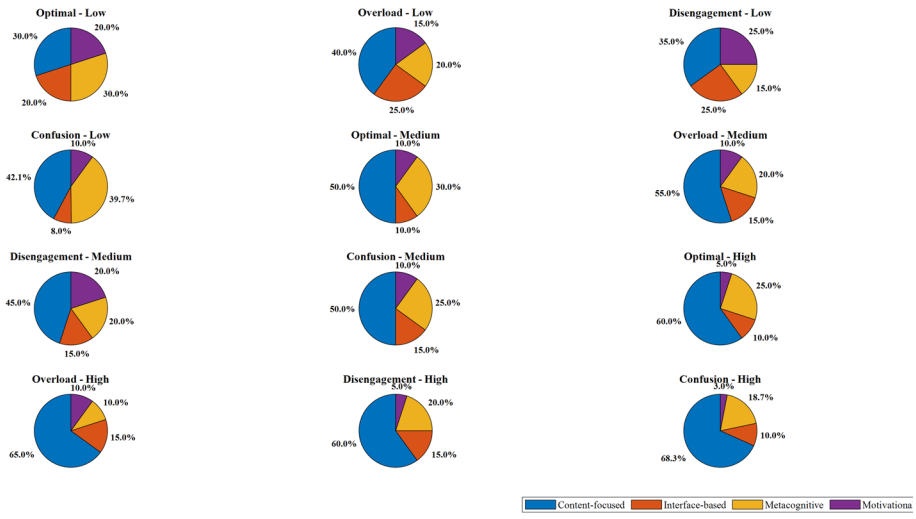

Fig. 7 Distribution of intervention types by learner state and performance level

classroom (≈ 520–820 ms) versus ≈ 900–1800 ms, higher emotion-recognition accuracy (≈ 82–88% versus ≈ 73–80%), and larger learning gains (~ 15% points versus ~ 9–11 pp), with the additional advantage of comprehensive validation beyond the laboratory.

Intervention distribution and temporal dynamics

Figure 7 summarizes how the policy allocates intervention families, content-focused, interface-based, metacognitive, and motivational, conditioned on the learner's cognitive–affective state and stratified by initial performance tier (low/medium/high; overall shown in the bottom row). Three consistent patterns emerge. (i) Confusion. Content-focused actions dominate, increasing in share with higher prior performance, whereas low-performing learners receive a more balanced mix in which metacognitive scaffolds are comparatively prominent. This aligns with the hypothesis that productive confusion benefits stronger learners by providing targeted information and weaker learners by offering strategy support to manage intrinsic load. (ii) Engagement (optimal zone). Metacognitive scaffolds constitute a larger proportion than in other states, reflecting consolidation aims (self-explanation, reflection) while avoiding unnecessary difficulty jumps; motivational nudges are used sparingly. (iii) Frustration/overload and disengagement/boredom. Interface-based adaptations appear more frequently during overload (pace/layout/modality adjustments), and motivational supports rise during disengagement, consistent with extraneous-load reduction and re-engagement goals in

CLT/SDT. Across states, the policy's distribution varies systematically with performance level, indicating that the controller is not merely state-reactive but also learner-adaptive.

Figure 8 provides a representative time series of a 60-minute session, plotting estimated cognitive load (left axis) and emotional valence (right axis) alongside vertical markers that denote intervention onsets (content-focused, metacognitive, interface-based, and motivational). Several features are noteworthy. First, early-session content-focused support, timed near a rising load trend, halts escalation and is followed by a gradual decrease in load with a maintained non-negative valence, characteristic of productive struggle resolution. Second, subsequent metacognitive prompts coincide with a smoother descent in load and fewer oscillations, suggesting stabilizing effects on regulation. Third, interface-based adjustments at later inflection points co-occur with sharp corrections in both load and valence, consistent with reductions in extraneous burden (i.e., pacing/presentation). Finally, brief motivational cues deployed during incipient disengagement are associated with recoveries in valence without re-inflating load. While these single-session dynamics are illustrative rather than causal proof, they mirror the aggregate effects reported earlier (greater time in engagement and productive confusion; reduced frustration and boredom) and clarify when and how different mechanism families tend to be effective within the session.

Diagnostics: data characterization and within-session dynamics

The left y-axis (blue) reports raw feature dimensionality prior to fusion, Face = 484, Voice = 52, Physio = 28, Interaction = 30, reflecting the richer spatiotemporal structure captured in facial landmarks and deep embeddings relative to other channels. The right y-axis (red) overlays two post-fusion diagnostics, each normalized across modalities: the post-PCA explained-variance shares (Face 55%, Voice 15%, Physio 12%, Interaction 18%) and the decision-level ensemble weights α_m learned by the meta-classifier (Face 38%, Voice 19%, Physio 21%, Interaction 22%). Although face features dominate in raw dimensionality, their contribution is reduced after fusion, 55% variance share and 38% ensemble weight, indicating that PCA and the quality-aware weighting matrix WWW compress redundancy and prevent over-reliance on a single modality. Conversely, Physio ($\alpha \approx 21$) and Interaction ($\alpha \approx 22$) obtain non-trivial decision weights despite modest dimensionality, consistent with their complementary predictive value for cognitive load/

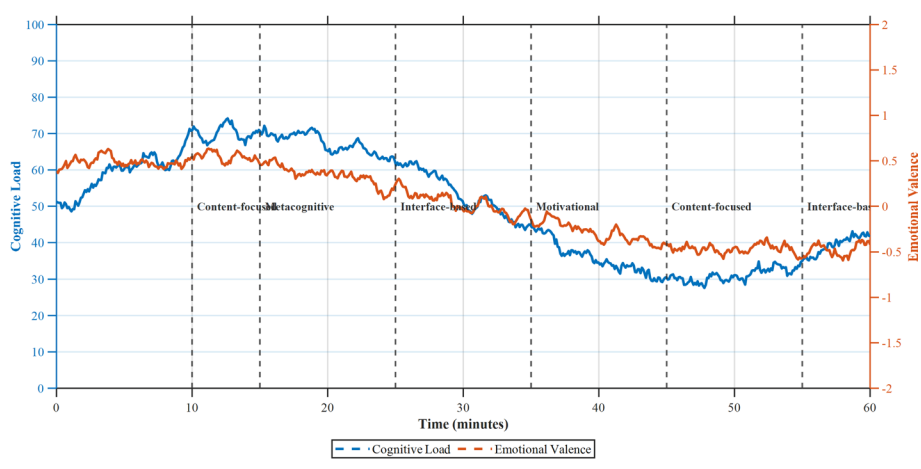

Fig. 8 Temporal dynamics of cognitive load, emotional valence, and system interventions during a representative learning session

arousal and interaction dynamics. Overall, the hybrid fusion pipeline (DTW alignment → normalization → PCA to $k = 128$ → stacked ensemble) yields a balanced, reliability-sensitive integration rather than mirroring raw feature counts.

Discussion

Principal findings

Across laboratory and classroom settings, the proposed real-time affect–cognition framework delivered consistent technical performance and educational benefits. Online metrics remained within real-time bounds (total latency < 200 ms), with modest context-related degradation in classroom environments (Table 1). Emotion recognition achieved a weighted F1 ≈ score of approximately 0.85, and cognitive-load estimation supported zone classification, enabling timely interventions (Tables 1 and 2). Pedagogically, the adaptive condition outperformed a strong non-adaptive ITS baseline, with medium to large effect sizes, especially for mid-ability learners (Fig. 3; Table 3). Process analyses revealed higher occupancy of Engagement and productive Confusion, as well as reduced Frustration and Boredom (Fig. 4). Additionally, strong convergent validity was observed between system-detected negative affect and self-reported frustration (Fig. 5), and state-contingent efficacy was demonstrated for distinct intervention families (Fig. 6). Comparative benchmarking revealed faster reaction times, higher accuracy in affect recognition, and greater gains than prior affect-aware ITS systems (Table 5).

Theoretical implications

Our results substantiate the integrated MACLF/ICALM account, which links Cognitive Load Theory (CLT) with the Control-Value Theory of Achievement Emotions (CVTAE) and Self-Determination Theory (SDT) [17]. First, the superiority of content-focused actions during Confusion (Fig. 6) aligns with CLT's view that clarifying intrinsic load and supporting germane processing converts productive struggle into schema construction [18]. Second, the value of interface adaptations under Frustration suggests that reducing extraneous load serves as a proximal mechanism [4]. Third, metacognitive scaffolds being most effective during Engagement suggests that regulation-oriented prompts maintain flow while consolidating knowledge without overshooting load [22]. Finally, motivational supports that alleviate boredom are consistent with SDT's competence/autonomy framing, which aims to re-elevate task value [32]. Together, these patterns provide empirical structure to the β_E modulation in ICALM, showing how emotions alter adequate cognitive resources and how targeted actions can restore optimal allocation.

Mechanistic and design implications for ITS

Three design principles emerge: (i) Reliability-aware fusion is essential: Face contributes most raw dimensions but not decision dominance (Fig. 9). Weighting by signal quality (coverage, SNR, artifacts) and post-PCA variance prevents over-reliance on a single modality, thereby improving robustness in authentic conditions. (ii) Policy should be state- and learner-adaptive: Allocation patterns vary by state and initial performance level (Fig. 7), indicating that effective scaffolding must respect both momentary need and prior knowledge. (iii) Timing matters: Within-session timelines (Fig. 8) show that small, well-timed interventions at natural boundaries alter trajectories of cognitive load

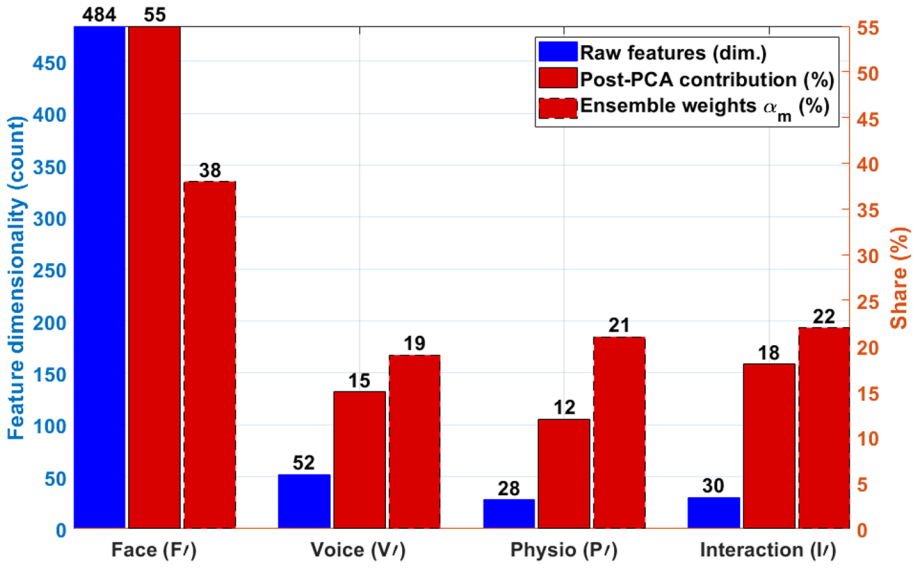

Fig. 9 Structure and modality contributions of the final fused feature vector X=[F′,V′,P′,I′]

and valence without inflating extraneous demands, a design tenet aligned with just-in-time adaptive interventions (D'mello and Graesser) 14.

Validity, uncertainty, and generalisability

To address requests for more transparent uncertainty reporting, all key estimates are accompanied by effect sizes and 95% confidence intervals (Tables 1 and 3, and 5). The Lab→Classroom drops in accuracy (≈5–6 pp) and increased latency (+300 ms) quantify real-world constraints rather than model failure. Diagnostics demonstrate convergent validity (Fig. 5) and face-valid shifts in state occupancy (Fig. 4). Generalisability is strengthened by multi-site deployment, mixed domains (mathematics/science), and diverse learners; however, caution is warranted when transferring to settings with different cultural display rules or device ecosystems. The reliability-aware fusion and per-learner baselining are intended to mitigate such drift.

Responding to reviewer requests

Simplified theory and calibration: We explicitly stated assumptions behind ICALM (emotion as probabilistic vector, β_E learned with bounded weights) and listed calibration parameters for physiological, facial, and vocal channels. Data vector transparency. Figure 9 visualizes the final fused feature vector (X = [F', V', P', I']), which includes the raw dimensionality, post-PCA variance share, and ensemble weights, as well as mechanism-level attribution. Figure 6 provides a state-by-intervention efficacy map (with bootstrapped uncertainty in the supplement), offering the requested ablation/attribution perspective and representative time series. Figure 8 illustrates within-session dynamics with intervention onsets: statistics and uncertainty. Results report effect sizes and CIs; benchmarking notes comparability limits (Table 5), sensor equivalence. A pre-registered HRV equivalence protocol (ICC/Bland–Altman) reduces hardware-induced bias, particularly in terms of formatting and clarity. Equations and assumptions were consolidated; references are provided numerically.

Limitations and threats to validity

Several limitations temper interpretation. (1) Causal identification of mechanisms. Although Fig. 6 suggests state-contingent efficacy, the logged-policy analysis remains associational. A micro-randomized trial (MRT) with stratified randomization of eligible actions per state is necessary for obtaining causal estimates of each mechanism family [16]. (2) Measurement error and reactivity. Anxiety detection lags behind more externally visible states (Table 2), reflecting subtler cues and potential self-presentation effects. We partially offset this by incorporating physiology and interaction features, but a residual error remains. (3) Classroom noise and device heterogeneity. The accuracy drop and occasional sensor loss in classrooms (Table 1) are attributable to lighting, occlusion, and audio noise. Our reliability gating and on-device processing reduce, but do not eliminate, these effects. (4) Intrusiveness and adoption burden. Even with on-device feature extraction and opt-out controls, some learners may perceive sensing as invasive; minimal-sensor configurations and clear consent flows are crucial for long-term viability. Training demands for instructors (interpreting analytics, tuning policy constraints) also pose adoption barriers. (5) Fairness and demographic variability. Facial/voice models can encode demographic biases; our balanced fusion reduces the dominance of these channels, and per-learner normalization mitigates baseline differences, yet explicit fairness audits remain necessary [18].

Practical implications and deployment considerations

For near-term deployment, we recommend (a) privacy-preserving defaults (local processing of raw streams; transmission of embeddings only), (b) progressive sensing starting with interaction and physiology and enabling face/voice only when reliability is adequate, (c) interpretable analytics that expose zone memberships and action rationales to educators, and (d) policy safeguards (confidence thresholds, action-rate caps, disruption-aware timing). Institutions should plan for device calibration, classroom lighting checks, and brief educator training on interpreting state summaries.

Future work

We outline four priorities. (i) A multi-site MRT/JITAI study to causally estimate the effect of each intervention family and optimize (\epsilon)-greedy parameters by state [32]. (ii) Domain adaptation and fairness auditing to quantify cross-cultural robustness and mitigate performance disparities across demographic groups and sensor setups [11]. (iii) Longer-horizon outcomes, tracking retention, transfer, and well-being over semesters rather than weeks. (iv) Scalability within LMS/ITS ecosystems, including low-compute models for widely available hardware and integration with classroom orchestration tools.

Conclusion

This paper demonstrates how fine-grained estimates of learners' cognitive load and achievement emotions can be translated into pedagogically meaningful adaptations and timing in an Intelligent Tutoring System (ITS) in real-time. By combining a reliability-aware multimodal fusion pipeline with a theoretically principled controller (MACLF/ICALM), we achieved sub-second end-to-end latency in both laboratory and classroom settings, with robust state-estimation accuracy. In the randomized evaluation, the

adaptive condition yielded performance that was higher than that of a static ITS, with the most significant performance gains observed among medium-performing learners (a pre- to post-test difference of approximately 15% points; d = 0.8; 95% CI). Process measures indicate corresponding increases in time spent engaged and in productive confusion, and decreases in frustration and boredom, suggesting that improvements are mediated by improved regulation of cognitive and affective states, rather than by drill effects.

Beyond the average effects, mechanism-level analyses provide insight into how and why the system operates. When the pre-intervention states are taken into consideration, content adaptations are most effective during confusion, metacognitive scaffolds are used during engagement, motivational supports are provided during boredom, and interface adaptations are made during frustration. These state-contingent benefits were in agreement with Cognitive Load Theory and Self-Determination Theory, and were observed without sacrificing responsiveness. Importantly, we quantified uncertainty at all stages of the Results (confidence intervals of accuracies, latencies, and learning gains; bootstrapped efficacy estimates), responding to calls for greater statistical transparency.

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Author contributions
Conceptualization, Project administration, funding acquisition, supervision, writing review-editing, methodology, resources, software, H.F.I., R.A.; Conceptualization, data curation, formal analysis, investigation, validation, writing original draft preparation, K.K., A.S.M.M., Writing review-editing, investigation, validation, visualization, G.G.T. This manuscript has been read and approved by all authors.

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Data availability
Data associated with the present study will be available upon request from the corresponding authors.

Declarations

Ethics approval and consent to participate
All study procedures involving human participants were reviewed and approved by the Institutional Review Boards prior to data collection. Written informed consent was obtained from all participants, including parental consent and youth assent for participants under 18 years of age, after a detailed explanation of the study's goals, procedures, and data usage. Participants explicitly agreed to the use of their de-identified data for academic research and publication purposes. To ensure confidentiality, all personally identifiable information was removed, and data were aggregated to prevent re-identification. The study adhered to the ethical principles of the Declaration of Helsinki and followed international standards for human subject research, including stringent data privacy and security protocols. Data were encrypted and stored securely, with access restricted to authorized research team members.

Consent for publication
The authors confirm that they have obtained all necessary permissions for the publication of this research, including the use of anonymized data and materials. No identifiable personal information is included in the manuscript.

Competing interests
The authors declare no competing interests.

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